Hydrostatic Forces

Friday, October 24, 2025 9:18 AM

Total surface forces: $\dfrac{\sum \partial \vec{F}_s}{\partial x \partial y \partial z} = -\nabla p$

Total body forces: $\dfrac{\sum \partial \vec{F}_b}{\partial x \partial y \partial z} = -\rho g \hat{k}$

Newton's 2nd law applied to fluid element:

$$\underbrace{\sum \delta \vec{F}}_{\sum \delta \vec{F}_s} = \underbrace{\delta m}_{\sum \delta \vec{F}_b} \cdot \vec{a}$$

$$\underbrace{-\nabla p\, \delta x \delta y \delta z}_{\sum \delta \vec{F}_s} - \underbrace{\rho g \delta x \delta y \delta z\, \hat{k}}_{\sum \delta \vec{F}_b} = \underbrace{(\rho \delta x \delta y \delta z)}_{\delta m} \vec{a}$$

general eqn of motion for fluids with no shear forces

$$\boxed{-\nabla p - \rho g \hat{k} = \rho \vec{a}}$$

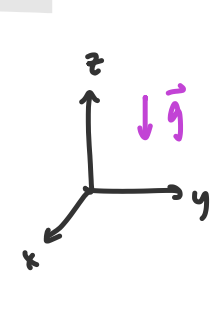
$-\nabla p$: net pressure force (per unit volume)

$\rho g \hat{k}$: net body force (per unit volume)

$\rho \vec{a}$: mass times acceleration (per unit volume)

pressure variation for fluid at rest

$\vec{a} = 0$



$$-\nabla p - \rho g \hat{k} = \cancel{\rho \vec{a}}^{\,0}$$

$$\nabla p = -\rho g \hat{k}$$

$$\Big(\underbrace{\frac{\partial p}{\partial x}}_{0} \hat{i} + \underbrace{\frac{\partial p}{\partial y}}_{0} \hat{j} + \underbrace{\frac{\partial p}{\partial z}}_{-\rho g} \hat{k}\Big) = -\rho g \hat{k}$$

$p = p(z)$

$$\boxed{\frac{dp}{dz} = -\rho g} \leftarrow \text{fluids at rest}$$

$$\int_{p_1}^{p_2} dp = -\int_{z_1}^{z_2} \rho g \, dz$$

ρ = const. : incompressible

$\rho \neq$ const. : compressible

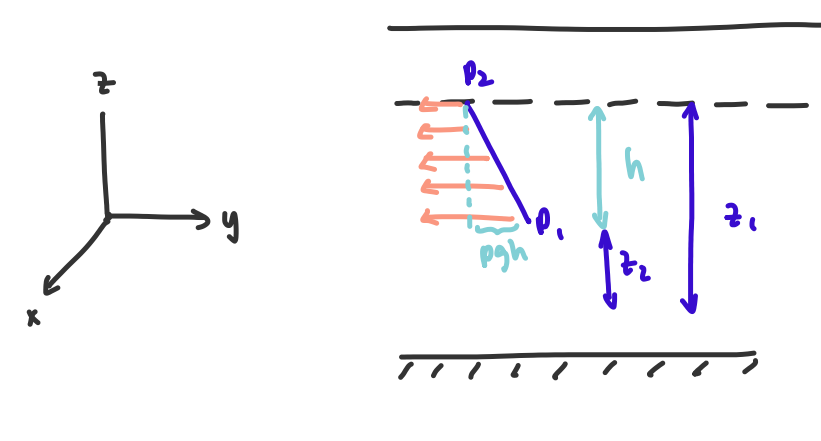
g = const.

Incompressible fluid at rest: ρ = const. (e.g. liquids in general)

$$\int_{p_1}^{p_2} dp = -\int_{z_1}^{z_2} \rho g \, dz \Rightarrow \int_{p_1}^{p_2} dp = -\rho g \int_{z_1}^{z_2} dz \Rightarrow p_2 - p_1 = -\rho g (z_2 - z_1)$$

$$\boxed{\begin{aligned} p_2 &= p_1 - \rho g h \\ p_1 &= p_2 + \rho g h \end{aligned}}$$

$h = z_2 - z_1$



$$p_1 - p_2 = \rho g h$$

$$= \rho g h (1) \leftarrow \text{weight of the fluid with height} = h \text{ and unit cross section}$$

(height, volume)

Compressible fluid at rest: $\rho \neq$ const. (e.g. gases)

$$\frac{dp}{dz} = \underbrace{-\rho g}$$

$\gamma = \rho g$: specific weight (weight per unit volume)

$(kg/m^3)(N/kg) = N/m^3$

$\rho_{gases} \ll \rho_{liquids}$

$(\rho g_{gases}) \ll (\rho g_{liquids})$

$$\left|\left(\frac{\partial p}{\partial z}\right)_{gases}\right| \ll \left|\left(\frac{\partial p}{\partial z}\right)_{liquids}\right|$$

Δh small : $\rho \approx$ const.

Δh large : $\rho = \rho(z) \Rightarrow$ $\boxed{\int_{p_1}^{p_2} dp = -\int_{z_1}^{z_2} \rho g \, dz}$ (ρ known)

Ex. Pressure elevation relationship for isothermal perfect gas

$p = \rho R T$

$$\frac{dp}{dz} = -\rho g \Rightarrow \frac{dp}{dz} = -\frac{p}{RT} g$$

$\rho = \dfrac{p}{RT}$

$$\int_{p_1}^{p_2} \frac{dp}{p} = \int_{z_1}^{z_2} \frac{g}{RT} dz$$

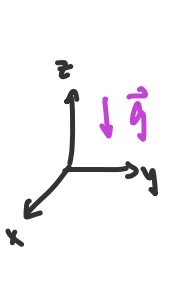
$\vdots$

$$p_2 = p_1 e^{-\frac{g}{RT}(z_2 - z_1)}$$

Summary

$-\nabla p + \rho \vec{g} = \rho \vec{a}$ — no shear

$-\nabla p - \rho g \hat{k} = \rho \vec{a}$ — no shear, $\vec{g} = -g\hat{k}$



$\dfrac{dp}{dz} = -\rho g \Rightarrow p = p(z)$ — no shear, $\vec{g} = -g\hat{k}$, fluid at rest

Incompressible fluid at rest: $p_2 = p_1 + \rho g h$

Compressible fluid at rest: if Δh small, $\rho \approx$ const. over h

if Δh large, $\int dp = \int \rho g \, dz$ (ρ should be known)

Ex. For an incompressible fluid at rest, grav. field in z dir, show that the force is horizontal

ρ: const.

$\vec{a}$: 0

$\vec{g} = -g\hat{k}$

$$-\nabla p - \rho g \hat{k} = \cancel{\rho \vec{a}}^{\,0} \Rightarrow \nabla p = -\rho g \hat{k}$$

$$\frac{dp}{dz} = -\rho g \Rightarrow \int dp = -\int \rho g \, dz$$

(const., const.)

$$p = -\rho g z + C$$

Boundary condition: $p = p_{atm}$ at $z = z_s$

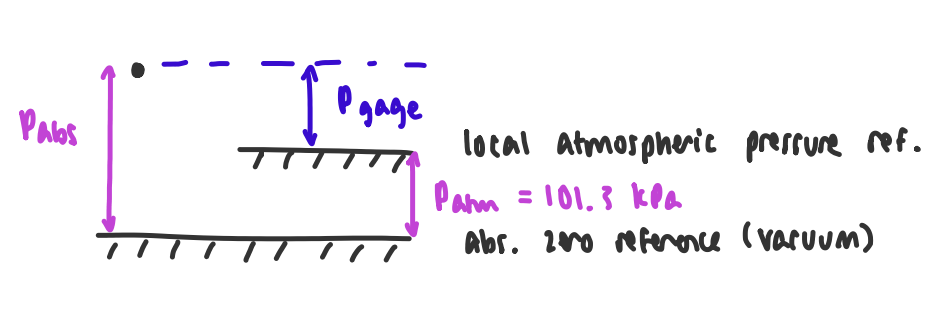
$$p_{atm} = -\rho g z_s + C$$

$$z_s = \frac{C - p_{atm}}{\rho g} \Rightarrow z \text{ is constant}$$

(const., const., const., const.)

Measurement of pressure

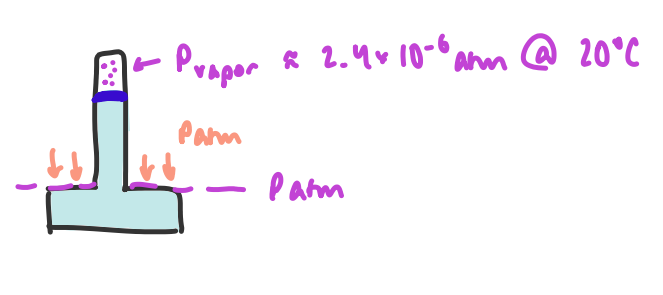
pressure must be stated wrt. a reference level



$$p_{abs} = p_{gage} + p_{atm}$$

Manometers

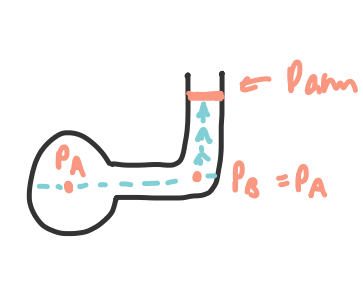
mercury barometer



$$p_A = p_{atm} - \rho g h = p_{vapor}$$

$$p_{atm} = \rho_{Hg} g h$$

piezometer



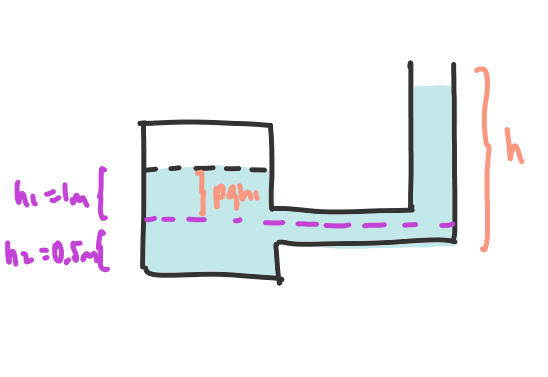
$p_A - \rho g h_1 = p_{atm}$ (abs. pressure, abs. pressure)

$p_A - \rho g h_1 = 0$ (gage pressure, gage pressure)

Ex. Closed tank contains air and water, connected with piezometer.

$h_1 = 1m$, $h_2 = 0.5m$.

If pressure gage reading of compressed air is 10 kPa, determine height (h) of water in piezometer ($\rho = 1000 \, kg/m^3$ for water, $g = 10 \, m/s^2$)

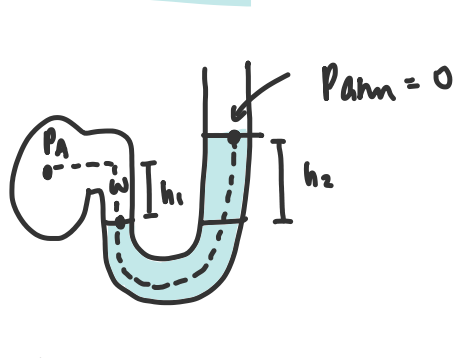


$$p + \rho g h_1 - \rho g h = 0$$

$$10000 + 1000(10)(1) - 1000(10)(h) = 0$$

$$h = 2m$$

U-tube manometer

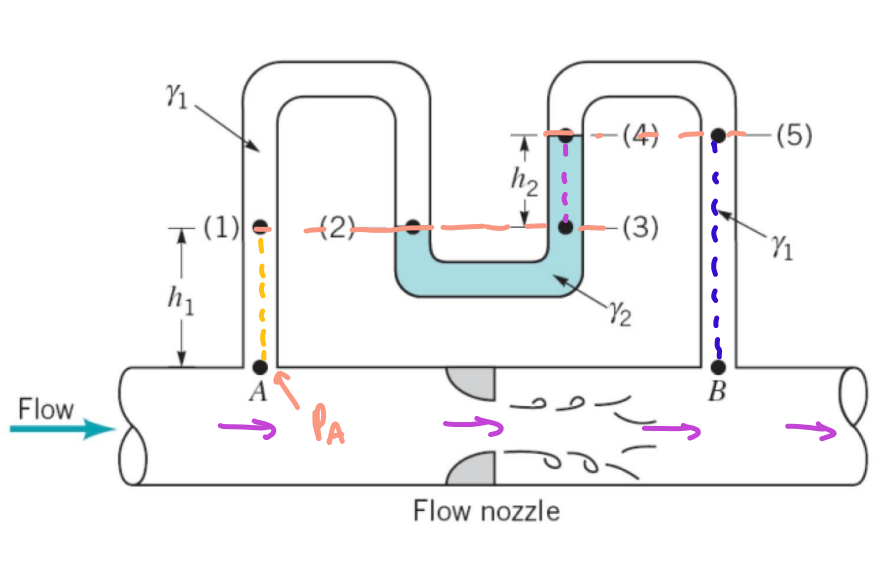


$$p_A + \gamma_1 h_1 - \gamma_2 h_2 = 0$$

$$p_A = \gamma_2 h_2 - \cancel{\gamma_1 h_1} \quad \text{when A has gas, } \gamma_1 \ll \gamma_2$$

$$p_A = \gamma_2 h_2$$

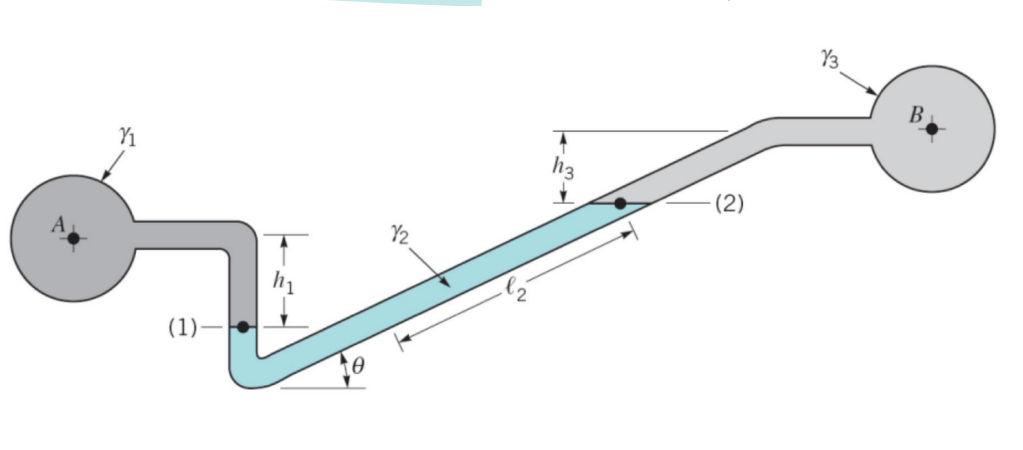
Ex. Determine eqn. of $p_A - p_B$ along pipe in terms of specific weight of flowing fluid γ_1 in the pipe, specific weight of gage fluid γ_2, and various heights.



$$p_A - \underline{\gamma_1 h_1} - \underline{\gamma_2 h_2} + \underline{\gamma_1 (h_1 + h_2)} = p_B$$

$$p_A - p_B = \gamma_1 h_1 + \gamma_2 h_2 - \gamma_1 (h_1 + h_2)$$

Inclined tube manometer — ideal to measure small differences in gas pressure



$$p_A + \gamma_1 h_1 - \gamma_2 \ell_2 \sin\theta - \gamma_3 h_3 = p_B$$

$$p_A - p_B = \gamma_2 \ell_2 \sin\theta + \cancel{\gamma_3 h_3} - \cancel{\gamma_1 h_1}$$

If A and B contains gas:

$$p_A - p_B = \gamma_2 \ell_2 \sin\theta$$

$$\ell_2 = \frac{p_A - p_B}{\gamma_2 \sin\theta}$$

as $\theta \to 0$, $\ell_2 \to \infty$

Hydrostatic forces on submerged surfaces

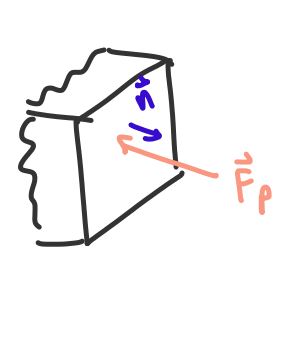
① $\vec{F}_p \perp$ surface

② No shear stresses

③ For incompressible fluids, pressure changes linearly w/ z

Hydrostatic forces acting on planar surfaces

method 1: integration

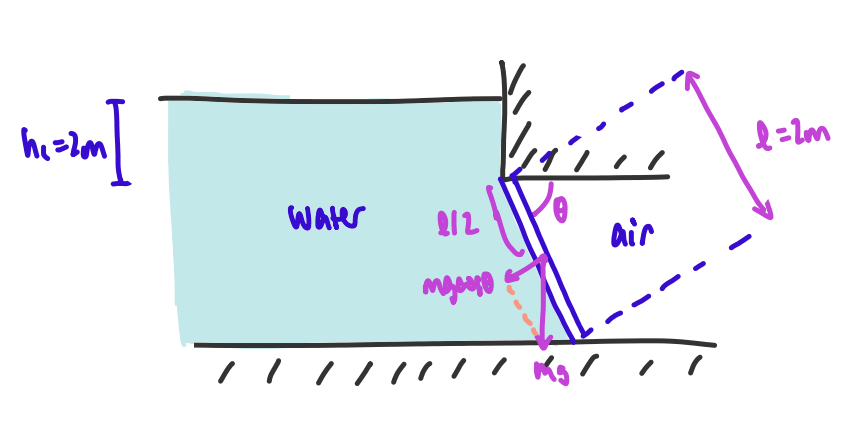


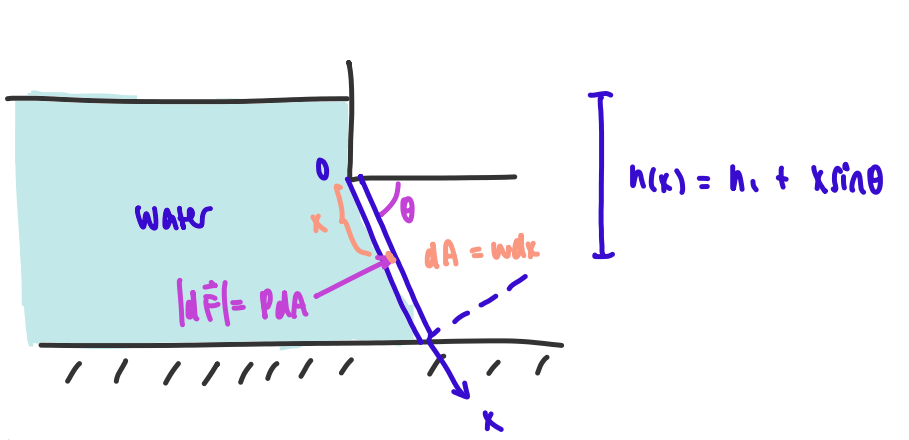
$$d\vec{F}_p = -p \, dA \, \vec{n}$$

$$\vec{F}_p = \iint_{Area} d\vec{F}_p$$

Ex. find mass of gate such that it will open when water level exceeds 2m.

$\rho_{water} = 1000 \, m^3/kg$





closing moment:

$$M_{closing} = mg\cos\theta \left(\frac{\ell}{2}\right) = mg(0.6)\frac{2}{2} = 0.6mg$$

opening moment:

$$M_{opening} = \iint_A x \underbrace{p dA}_{\text{moment on dA}}$$

$p = \rho g h(x)$ ← dA = wdx, w is constant

$$= \int_{x=0}^{\ell} x \rho g h(x) w dx$$

$$= \int_{x=0}^{\ell} x \rho g (h_1 + x\sin\theta) w dx$$

$$= \rho g w \int_0^{\ell} (x h_1 + x^2 \sin\theta) dx$$

$$= \rho g w \left[h_1 \frac{x^2}{2} + \sin\theta \frac{x^3}{3}\right]_0^{\ell}$$

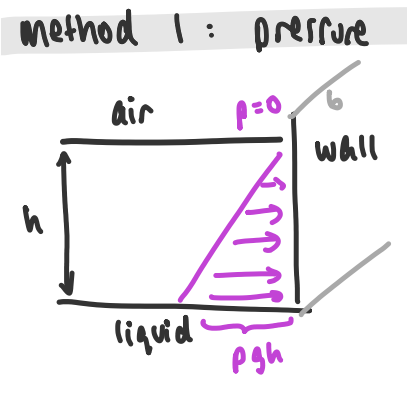
$$= \rho g w \left(\frac{18.4}{3}\right)$$

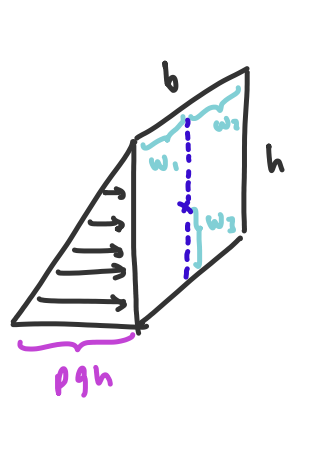
$$M_{opening} = M_{closing}$$

$$0.6mg = \rho g w \left(\frac{18.4}{3}\right)$$

$$m = \frac{\rho}{0.6} w \left(\frac{18.4}{3}\right) = \frac{1000}{0.6}(1.8)\left(\frac{18.4}{3}\right) = 18400 \text{ kg}$$

method 1: pressure prism concept ← only for rectangular regions





(pressure-area space)

Pressure prism

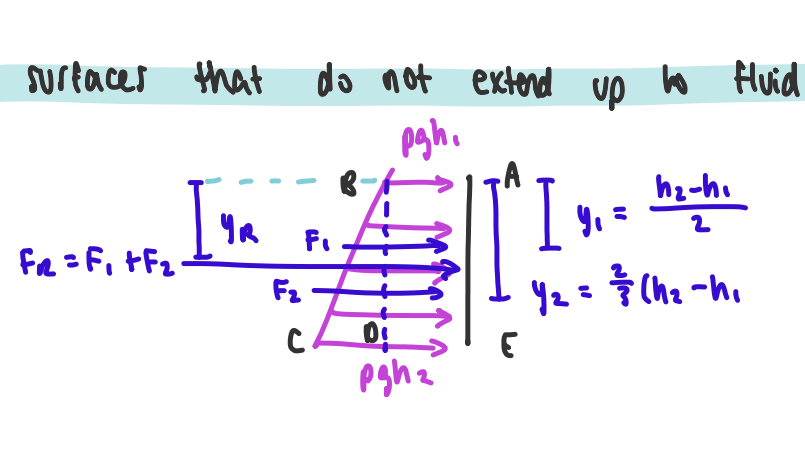
$$p_{av} = \frac{0 + \rho g h}{2} = \frac{\rho g h}{2}$$

$$F_R = p_{av} \cdot A = \frac{\rho g h}{2} \cdot hb = \text{volume of pressure prism} = \frac{1}{2} \rho g h \cdot hb$$

F_R: magnitude of resultant pressure force = volume of pressure prism

Location of application of $\vec{F}_R$ passes through centroid of pressure prism.

Note: surfaces that do not extend up to fluid surface



$$F_R = \text{volume}|_{ABCE} = \underbrace{\text{volume}|_{ABDE}}_{F_1} + \underbrace{\text{volume}|_{BCD}}_{F_2}$$

$$F_R \cdot y_R = F_1 y_1 + F_2 y_2 \Rightarrow y_R \text{ can be found}$$

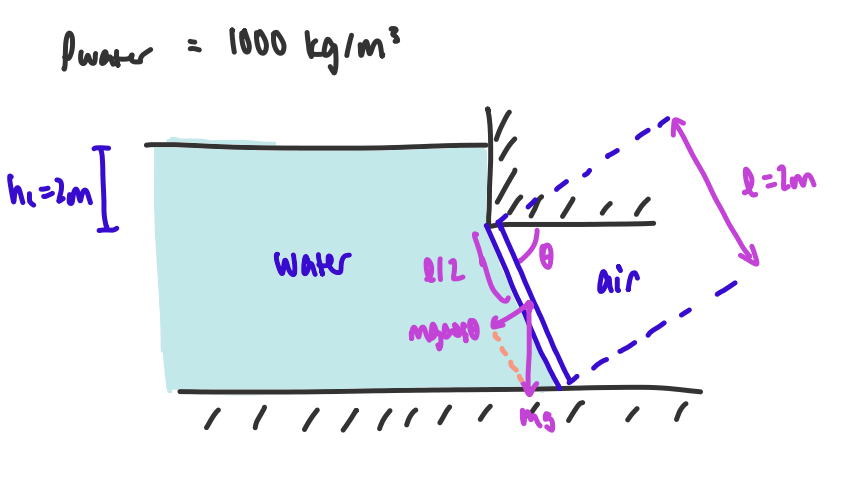
$y_1 = \frac{h_2 - h_1}{2}$, $y_2 = \frac{2}{3}(h_2 - h_1)$

* only gage pressure contributes to the resultant force, since p_{atm} cancels out on both sides

Important: arrows must be perpendicular to surface

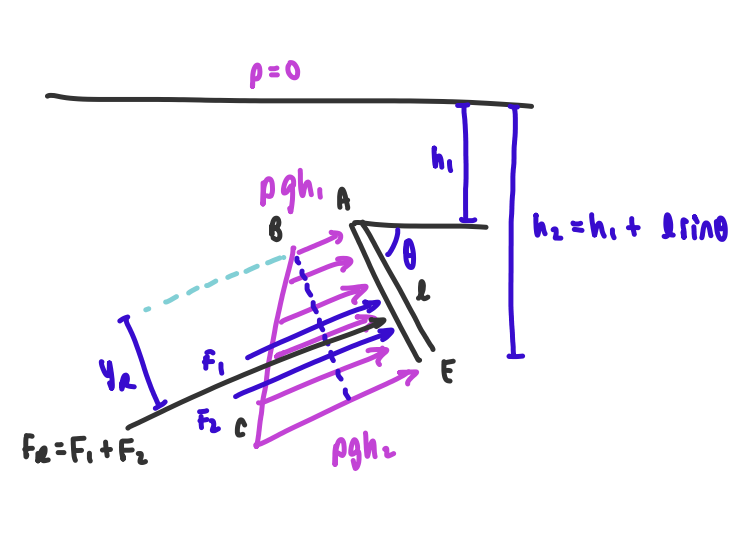
Ex. find mass of gate such that it will open when water level exceeds 2m.

$\rho_{water} = 1000 \, kg/m^3$



$M_{closing} = mg\cos\theta \frac{\ell}{2}$

$M_{opening} = ?$



$$F_R = \text{Volume}|_{ABCE} = \underbrace{\text{Volume}|_{ABDE}}_{F_1} + \underbrace{\text{Volume}|_{BCD}}_{F_2}$$

$$F_1 = \text{volume}|_{ABDE} = \rho g h_1 \ell w = 72000 \text{ N}$$

$$F_2 = \text{volume}|_{BCD} = \rho g (h_2 - h_1) \frac{\ell}{2} w = \frac{\rho g \ell^2 \sin\theta}{2} w = 28800 \text{ N}$$

$$F_R = F_1 + F_2 = 100\,800 \text{ N}$$

$$F_R \cdot y_R = F_1 \cdot \frac{\ell}{2} + F_2 \cdot \frac{2}{3}\ell \Rightarrow y_R = 1.085 \text{m}$$

$$M_{closing} = M_{opening} \Rightarrow mg\cos\theta \frac{\ell}{2} = F_R y_R$$

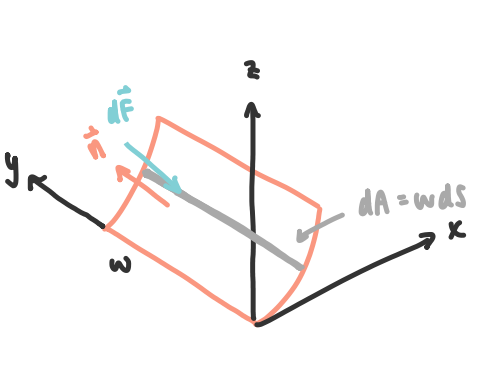
$$m(10)(0.6)\left(\frac{2}{2}\right) = (100800)(1.085)$$

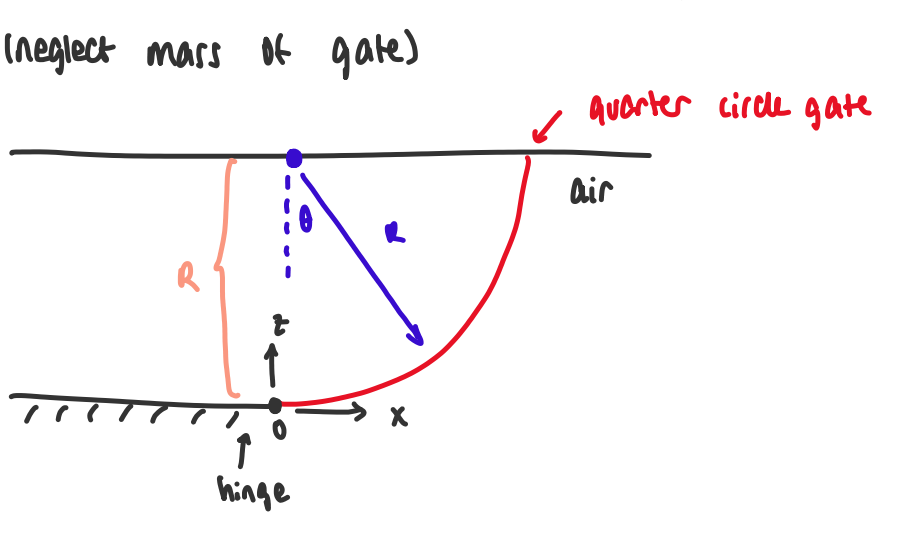
$$m = 18400 \text{ kg}$$

Pressure along curved surface: Integration methods

Method 1: integration into strips

Ex. find the horizontal force to keep the gate closed. (neglect mass of gate)

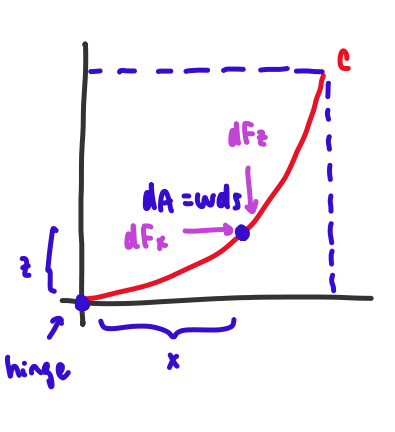




$$F_R = M_{water}$$

$$dM = x|dF_z| + z|dF_x|$$

$$M = \iint_{surface} dM = \iint_{surface} x|dF_z| + z|dF_x|$$



$$d\vec{F} = \underset{>0}{dF_x} \hat{i} + \underset{<0}{dF_z} \hat{k}$$

$$\|d\vec{F}\| = dF = \underbrace{p dA} = p w ds$$

dA = wds

$$dF_x = p w \sin\theta \, ds$$

$$dF_z = p w \cos\theta \, ds$$

parametrize C: $x, z \to \theta$

$x = R\sin\theta$

$z = R - R\cos\theta$ } $0 < \theta < \frac{\pi}{2}$

$$\vec{r}(\theta) = \underbrace{R\sin\theta}_{x} \hat{i} + \underbrace{(R - R\cos\theta)}_{z} \hat{k}$$

$$\vec{r}'(\theta) = R\cos\theta \hat{i} + R\sin\theta \hat{k}$$

$$\|\vec{r}'(\theta)\| = \sqrt{R^2\cos^2\theta + R^2\sin^2\theta} = R$$

$$ds = \|\vec{r}'(\theta)\| d\theta \Rightarrow ds = R d\theta$$

$$dF_x = p w \sin\theta \, ds$$

$$dF_z = -p w \cos\theta \, ds$$

$$p = \rho g R\cos\theta$$

$$M = \int_C x p w \cos\theta \, ds + \int_C z p w \sin\theta \, ds$$

$$= \int_{\theta=0}^{\pi/2} R\sin\theta \, p w \cos\theta \, R d\theta + \int_0^{\pi/2} (R - R\cos\theta) p w \sin\theta \, R d\theta$$

$$= \int_0^{\pi/2} R \underset{p = \rho g R\cos\theta}{p} w \sin\theta \, R d\theta = \int_0^{\pi/2} R^3 \underbrace{\rho g w}_{const.} \sin\theta\cos\theta \, d\theta = R^3 \rho g w \int_0^{\pi/2} \sin\theta\cos\theta \, d\theta$$

$$= R^3 \rho g w \int_0^{\pi/2} \frac{1}{2} \frac{\sin 2\theta}{2} 2 d\theta = \frac{R^3 \rho g w}{4} (-\cos 2\theta)\Big|_0^{\pi/2} = \frac{R^3 \rho g w}{2}$$

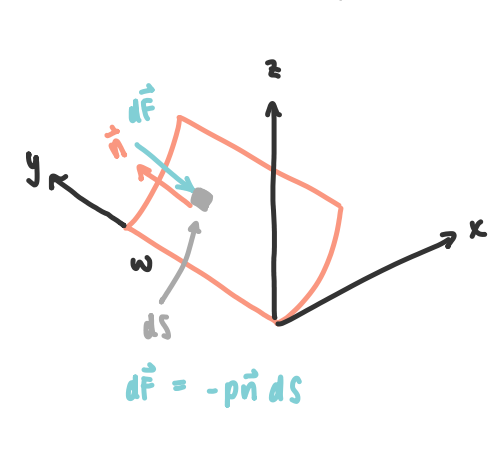
$$F \cdot R = M_{water}$$

$$F = \frac{M_{water}}{R} = \frac{R^3 \rho g w}{2} \cdot \frac{1}{R} = \frac{R^2 \rho g w}{2}$$

$$F = \frac{2^2 \cdot 1000 \cdot 10 \cdot 6}{2}$$

$$= 120\,000 \text{ N}$$

method 2: integration by dividing into patches



parametrize surface:

$x = R\sin\theta$, $z = R - R\cos\theta$, $y = y$, $0 \le \theta \le \frac{\pi}{2}$, $0 \le y \le w$

$$\vec{r}(\theta, y) = R\sin\theta \hat{i} + y\hat{j} + (R - R\cos\theta)\hat{k}$$

$$\vec{r}_\theta = R\cos\theta \hat{i} + R\sin\theta \hat{k}$$

$$\vec{r}_y = \hat{j}$$

$$\vec{r}_\theta \times \vec{r}_y = \begin{vmatrix} i & j & k \\ R\cos\theta & 0 & R\sin\theta \\ 0 & 1 & 0 \end{vmatrix} = -R\sin\theta \hat{i} + R\cos\theta \hat{k}$$

$$\|\vec{r}_\theta \times \vec{r}_y\| = R$$

$$\vec{n} = \frac{\vec{r}_\theta \times \vec{r}_y}{\|\vec{r}_\theta \times \vec{r}_y\|}$$

$$d\vec{F} = -p\vec{n} \, ds$$

$$= -p \frac{\vec{r}_\theta \times \vec{r}_y}{\|\vec{r}_\theta \times \vec{r}_y\|} ds$$

$$= (p\sin\theta \hat{i} - p\cos\theta \hat{k}) ds$$

$$d\vec{F}_x = p\sin\theta \, ds$$

$$d\vec{F}_z = -p\cos\theta \, ds$$

$$M_{opening} = M_y = \iint_S \underline{x|dF_z|} + \underline{z|dF_x|}$$

$$= \iint_S \left(\underline{R\sin\theta \, p\cos\theta} + \underline{(R - R\cos\theta) p\sin\theta}\right) \underset{\|\vec{r}_\theta \times \vec{r}_y\| dy d\theta = R dy d\theta}{ds}$$

$$= \int_0^{\pi/2}\int_0^w R\sin\theta \, p\cos\theta \, R dy d\theta + \int_0^{\pi/2}\int_0^w (R - R\cos\theta) p\sin\theta \, R dy d\theta$$

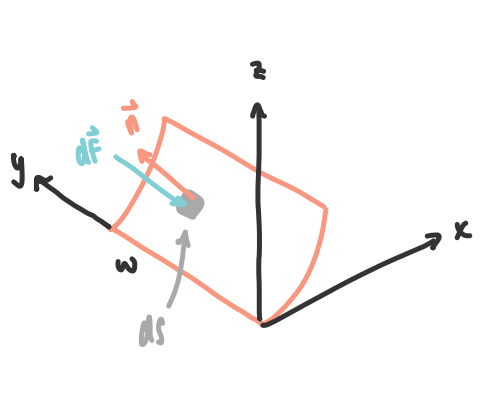
$$= \int_0^{\pi/2} R\sin\theta \, p\cos\theta \, R w \, d\theta + \int_0^{\pi/2} (R - R\cos\theta) p\sin\theta \, R w \, d\theta$$

$$= \int_0^{\pi/2} R^2 p\sin\theta \, w \, d\theta$$

$\vdots$

(same as before)

method 3: directly integrating with $\vec{M} = \vec{r} \times \vec{F}$



parametrize surface:

$x = R\sin\theta$, $z = R - R\cos\theta$, $y = y$, $0 \le \theta \le \frac{\pi}{2}$, $0 \le y \le w$

$$\vec{r}(\theta, y) = R\sin\theta \hat{i} + y\hat{j} + (R - R\cos\theta)\hat{k}$$

$$d\vec{F} = -p \, ds \, \vec{n}$$

$$d\vec{M}_0 = \vec{r} \times d\vec{F}$$

$$dM_{opening} = \|d\vec{M}_y\| = (\vec{r} \times \underset{-p ds \vec{n}}{d\vec{F}}) \cdot \hat{j} = [(\vec{r} \times \vec{n}) \cdot \hat{j}](-\underset{p = \rho g R\cos\theta}{p} ds)$$

$$\vec{r}_\theta \times \vec{r}_y = -R\sin\theta \hat{i} + R\cos\theta \hat{k}$$

$$\|\vec{r}_\theta \times \vec{r}_y\| = R$$

$$\vec{n} = \frac{\vec{r}_\theta \times \vec{r}_y}{\|\vec{r}_\theta \times \vec{r}_y\|}$$

$$\vec{r} \times \vec{n} = \begin{vmatrix} i & j & k \\ R\sin\theta & y & R - R\cos\theta \\ -\sin\theta & 0 & \cos\theta \end{vmatrix} = y\cos\theta \hat{i} - R\sin\theta \hat{j} + y\sin\theta \hat{k}$$

$$\|d\vec{M}_y\| = \underbrace{(\vec{r} \times \vec{n}) \cdot \hat{j}}_{-R\sin\theta} (-\underset{p = \rho g R\cos\theta}{p} \, ds)$$

$ds = \|\vec{r}_\theta \times \vec{r}_y\| d\theta dy$

$$dM_y = \rho g R^3 \sin\theta\cos\theta \, d\theta \, dy$$

$$M_y = \iint_S dM_y$$

$$= \int_0^{\pi/2}\int_0^w \rho g R^3 \sin\theta\cos\theta \, dy \, d\theta$$

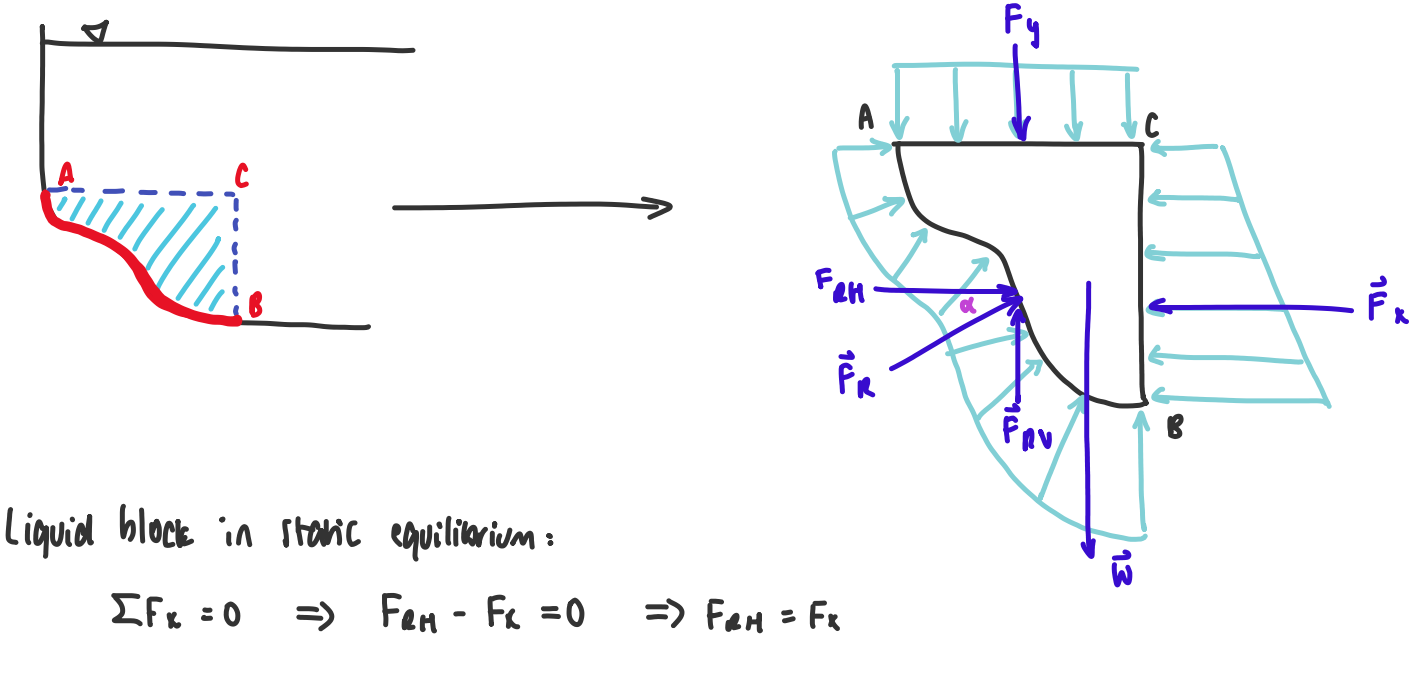
$$= \int_0^{\pi/2} \rho g R^3 \sin\theta\cos\theta \, w \, d\theta$$

$\vdots$

(rest is the same)

Curved surfaces: graphical method

Consider equilibrium of fluid volume enclosed by curved surface of interest and horizontal / vertical projection of surface passing through two ends of surface.



Liquid block in static equilibrium:

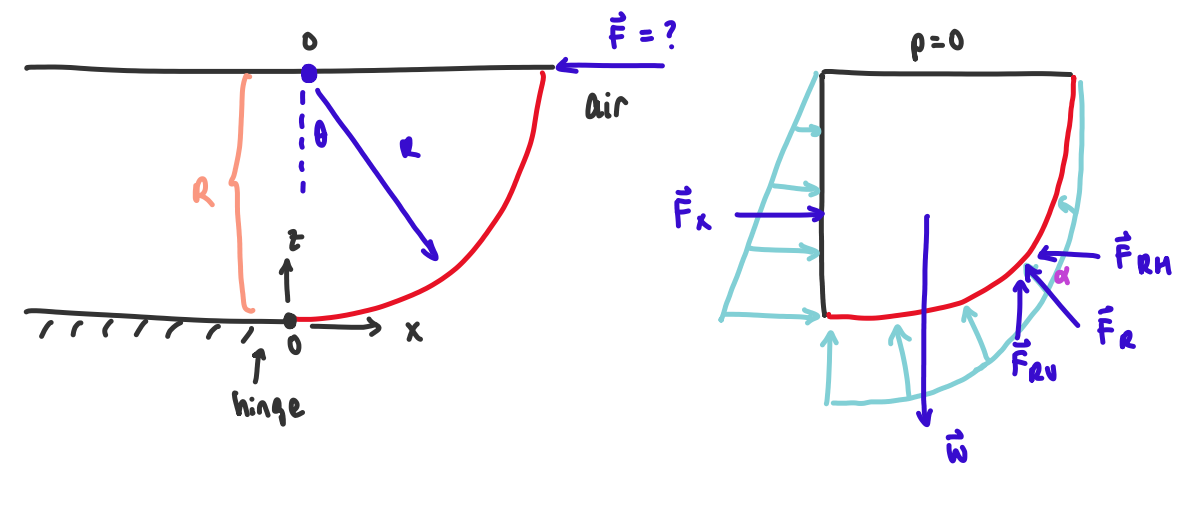
$$\sum F_x = 0 \Rightarrow F_{RH} - F_x = 0 \Rightarrow F_{RH} = F_x$$

$$\sum F_y = 0 \Rightarrow F_{RV} - F_y - W = 0 \Rightarrow F_{RV} = F_y + W$$

$$F_R = \sqrt{F_{RV}^2 + F_{RH}^2}$$

$$\alpha = \tan^{-1} \frac{F_{RH}}{F_{RV}}$$

Ex. Horizontal force F required to keep gate closed.



$$\sum F_x = 0 : F_{RH} = F_x = \frac{1}{2}\rho g R^2 w = \frac{1}{2}(1000)(10)(2^2)(6) = 120\,000 \text{ N}$$

$$\sum F_y = 0 : F_{RV} = W = \rho g \cdot \text{volume} = \rho g \frac{\pi R^2}{4} w = 1000 \cdot 10 \cdot \frac{\pi 2^2}{4} \cdot 6 = 60000\pi \text{ N}$$

$$F_R = \sqrt{F_{RH}^2 + F_{RV}^2} = 223\,451 \text{ N}$$

$$\alpha = \tan^{-1}\left(\frac{F_{RV}}{F_{RH}}\right) = 57.5°$$

for the gate, in equilibrium: $\sum M_{hinge} = 0 \Rightarrow F \cdot R = F_R \cdot R\cos\alpha \Rightarrow F = 120\,000 \text{ N}$